THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Full Syllabus (Chapters 1–10) | Class 6 Maths | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(b)** 21	Each term is the sum of the two terms before it.
2	**(d)** 90	At full hours the minute hand is at 12; each hour mark is 30°.
3	**(c)** 55	Reverse and add.
4	**(b)** 4.5	Divide by the value of one symbol.
5	**(b)** 3516	Apply the divisibility rule of 4.
6	**(c)** 40	Area $= \frac{1}{2} \times$ base $\times$ height.
7	**(b)** $\frac{5}{6}$	Convert to a common denominator first.
8	**(b)** outside the circle	Compare 9 cm with the radius 5 cm.
9	**(c)** 2	Along its two diagonals.
10	**(d)** -5	Same signs: add and keep sign; different signs: subtract.
11	**(a)** 3	Number of lines $= \frac{3\times2}{2}$.
12	**(a)** acute	Classify the turn.
13	**(d)** 5175	Arrange digits in descending and ascending order and subtract.
14	**(a)** 8	Divide the number by the scale.
15	**(b)** 8 and 15	Co-prime numbers have no common factor other than 1.
16	**(d)** 24	Half the product of the legs.
17	**(b)** $\frac{5}{12}$	Common denominator first.
18	**(c)** 4	Midpoint of AB.
19	**(b)** 72	$360^\circ \div$ number of sides.
20	**(b)** -7	Add the opposite.

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	Each term is the sum of the previous two (1); $1, 1, 2, 3, 5, 8, 13, 21, 34, 55$ (1).	1+1
22	Adjacent angles add up (1); $46 + 38 = 84^\circ$ (1).	1+1
23	47 each gets reversed and added (1); Palindrome reached: 121 after 1 step(s) (1).	1+1
24	Number of symbols = number $\div 5$ (1); Guava: 5 symbols; Mango: 2 symbols; Banana: 7 symbols (1).	1+1
25	Last two digits 16 give divisible by 4 (1); Last three digits 216: divisible by 8 (1).	1+1
26	Area $= \frac{1}{2} \times b \times h$ (1); $\frac{1}{2} \times 12 \times 9 = 54$ cm^2 (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	Sum $= 45 + 55 = 100$ (1); $100 = 10 \times 10$ (1); Two consecutive triangular numbers always add to a square number (1).	1+1+1
28	Every 90° is a quarter turn: N to E to S to W (1); 90° gives East (1); Type: right (1).	1+1+1
29	Repeat the largest-minus-smallest step (1); $3524 \to 3087 \to 8352 \to 6174$ (1); Number of steps $= 3$ (1).	1+1+1
30	(i) Banana: 40 (1); (ii) Apple: 10 (1); (iii) total $= 80$ (1).	1+1+1
31	Sum of digits at odd places and even places (1); Difference $= 11$ (1); Divisible by 11 (1).	1+1+1
	OR: Perimeter $= 2(15 + 8) = 46$ cm (1); Area $= 120$ cm^2 (1); Units must be written (1).	1+1+1
32	LCM of denominators $= 12$ (1); Sum $= \frac{17}{12}$ (1); Greater than 1 (1).	1+1+1
	OR: Draw AB = 6 cm (1); With A and B as centres and the same radius more than half of AB, draw arcs on both sides (1); Join the intersection points; the line meets AB at its midpoint, which is at 3 cm from A (1).	1+1+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	First person shakes 5 hands, next 4 new hands, ... down to 1 (1); Total $= 5+4+3+2+1$ (1); $= 15$ handshakes (1); This is the triangular number T_5 (2).	1+1+1+2
34	$\angle AOC = 110 - 30 = 80^\circ$ (1); $\angle AOB = 30^\circ$: acute (1); $\angle BOC = 110^\circ$: obtuse (1); $\angle AOC = 80^\circ$: acute (1); State the arms and vertex (1).	1+1+1+1+1
35	Step 1: 4311-1134=3177 (1); Step 2: result 6354 (1); Step 3: result 3087 (1); The numbers move toward the Kaprekar constant 6174 (2).	1+1+1+2
36	Tennis: 3 symbols; Cricket: 6 symbols; Kabaddi: 7 symbols (1); Total symbols $= 16$ (1); Total students $= 80$ (1); Write the key: 1 symbol = 5 students (2).	1+1+1+2
	OR: Cross out multiples of 2, 3, 5, 7 (1); Primes: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37 (1); Number of primes $= 12$ (1); Check by testing factors (2).	1+1+1+2
37	Area of first part $= 10 \times 6 = 60$ cm^2 (1); Area of second part $= 4 \times 5 = 20$ cm^2 (1); Total area $= 80$ cm^2 (1); Difference $= 40$ cm^2 (1); State the units (1).	1+1+1+1+1
	OR: Total =\frac{1}{4}+\frac{1}{3}= $7/12$ (1); Left $= 1 - 7/12 = 5/12$ (1); More on Tuesday (1); Check total plus left equals 1 (2).	1+1+1+2
38	Draw AB = 8 cm (1); Draw perpendiculars at A and B and mark 6 cm on each (1); Join the free ends (1); Draw both diagonals and measure: AC = BD (1); Diagonals bisect each other so OA = OC = AC/2 (1).	1+1+1+1+1
	OR: (a) 6 (1); (b) 6 (1); (c) 60° (1); (d) $4 \times 60 = 240^\circ$ (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Full Syllabus (Chapters 1–10) | Class 6 Maths | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(a)** 25	Two consecutive triangular numbers add up to a square number.
2	**(a)** 75	Add the adjacent angles.
3	**(a)** 9	Even: halve it. Odd: multiply by 3 and add 1.
4	**(a)** 20	Subtract the smaller number from the larger.
5	**(b)** 11 and 13	Twin primes are primes that differ by 2.
6	**(d)** 32	Find the side first.
7	**(b)** $\frac{2}{3}$	Compare using a common denominator.
8	**(d)** 5	Diagonals of a rectangle are equal and bisect each other.
9	**(b)** the same letter	Vertical-axis symmetry keeps it the same.
10	**(a)** -6	Right is positive, left is negative.
11	**(d)** 120	Triangular number $T_{15} = \frac{15\times16}{2}$.
12	**(b)** 270	Remaining turn $= 360^\circ -$ turn.
13	**(c)** 2	Keep reversing and adding.
14	**(d)** 75	Add all.
15	**(c)** 6	Check each number.
16	**(b)** 9	Breadth = area ÷ length.
17	**(c)** $3\frac{5}{6}$	Add the whole parts and fraction parts separately.
18	**(a)** 60	Triangle OAB is isosceles with OA = OB.
19	**(a)** 4	Order = $360 \div$ smallest angle.
20	**(c)** 8	Count the integers strictly between.

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	LCM of 4 and 6 is 12 (1); $\frac{5}{12}$ ($\frac{5}{12}$) (1).	1+1
22	Compare 4 cm with the radius 6 cm (1); P lies inside the circle (1).	1+1
23	Fold the figure so that the two halves coincide; There are 2 lines of symmetry (1); Lines pass through vertices/midpoints as per the shape (1).	1+1
24	5 yellow tokens and 3 red tokens (1); Cancel pairs: the value is 2 (1).	1+1
25	Each point joins to 7 others, and every line is counted twice (1); Lines $= \frac{8\times7}{2} = 28$ (1).	1+1
26	Hour hand points to 3 on the dial; each mark is 30° (1); Angle $= 90^\circ$ (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	Sides $= 360 \div 40 = 9$ (1); Lines of symmetry $= 9$ (1); Order of rotational symmetry equals the number of sides (1).	1+1+1
28	First two: -4 (1); Then add 5 (1); Result 1 (1).	1+1+1
29	$1 + 2 + \cdots + 3 = 6$ (1); $1^3 + \cdots + 3^3 = 1 + 8 + 27 = 36$ (1); $6^2 = 36$, so they match (1).	1+1+1
30	Draw a ray OA, place the centre of the protractor at O with the zero line along OA (1); Mark a point at 40° and join O to it (1); The angle is acute (1).	1+1+1
31	Follow the rule step by step (1); Sequence: 13, 40, 20, 10, 5, 16, 8, 4, 2, 1 (1); Steps $= 9$, maximum $= 40$ (1).	1+1+1
	OR: Values: 40, 70, 50 (1); Largest $= 70$, smallest $= 40$ (1); Difference $= 30$ (1).	1+1+1
32	Split the number into two factors repeatedly (1); $96 = 2 \times 2 \times 2 \times 2 \times 2 \times 3$ (1); Product check: 96 (1).	1+1+1
	OR: Legs 6 cm and 8 cm; area $= \frac{1}{2} \times 6 \times 8 = 24$ cm^2 (1); Perimeter $= 6 + 8 + 10 = 24$ cm (1); Units: cm and cm^2 (1).	1+1+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	After rising: $-80 + 30 = -50$ m (1); After diving: $-50 - 25 = -75$ m (1); Total distance travelled $= 30 + 25 = 55$ m (1); Below sea level (2).	1+1+1+2
34	Next term $= 8 + 13 = 21$ (1); Then 34 and 55 (2); Sum of first 5 terms $= 19$ (equals the term two places ahead minus 2: 21-2) (2).	1+2+2
35	The hour hand turns 30° every hour (1); Hours passed $= 4 - 1 = 3$, hour hand turn $= 90^\circ$ (1); Minutes passed $= 180$; the minute hand turns 3 full turns $= 1080^\circ$ (1); Type of the hour hand's turn: right (2).	1+1+1+2
36	121+232=353 (1); Palindrome (1); Start 29: 29, 121 (1); Palindrome 121 after 1 step(s) (2).	1+1+1+2
	OR: Count each value: 0: 1; 1: 1; 2: 3; 3: 3; 4: 2; 5: 4 (1); Most common: 5 (2); More than 2: 9 students (2).	1+2+2
37	3 and 4 have no common factor (1); Example: $60 \div 3 = 20$ and $\div 4 = 15$ (1); $60 \div 12 = 5$ (1); So both factors divide it and their product also divides (2).	1+1+1+2
	OR: Outer rectangle: 24 m by 19 m (1); Outer area $= 456$ m^2 (1); Garden area $= 300$ m^2 (1); Path area $= 156$ m^2 (1); State units (1).	1+1+1+1+1
38	After pouring: $3\frac{5}{6}$ litres (1); Subtract $\frac{1}{4}$ (1); Left $3\frac{7}{12}$ litres (1); Check with improper fractions (2).	1+1+1+2
	OR: P: inside (1); Q: on (1); R: outside (1); Inside or on the circle: 2 (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Full Syllabus (Chapters 1–10) | Class 6 Maths | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(d)** 256	16×16.
2	**(b)** 6	Each pair of rays makes an angle.
3	**(c)** 22	Add all the digits.
4	**(b)** 2	Count how many times each value appears.
5	**(c)** 5	$72 = 2 \times 2 \times 2 \times 3 \times 3$
6	**(d)** 1200	Cost = perimeter × rate.
7	**(c)** $\frac{2}{3}, \frac{3}{4}, \frac{5}{6}$	Use a common denominator.
8	**(d)** 4	Subtract the radii.
9	**(a)** 7	Sides = lines of symmetry.
10	**(b)** -9	Add the rise.
11	**(d)** 45	Triangular numbers: term n is $1 + 2 + \cdots + n$.
12	**(b)** 145	Inner and outer scales add up to $180°$.
13	**(c)** 31	Add the two cells below.
14	**(b)** 6	Count the values bigger than the given number (not equal to it).
15	**(c)** 3	The next un-crossed number after 2 is 3.
16	**(c)** 300	Divide the floor area by the area of one tile.
17	**(a)** 3	Reduce the fraction.
18	**(d)** equal and perpendicular bisectors of each other	Property
19	**(d)** Z	Z fits after a half turn but cannot be folded.
20	**(b)** 0	$n + (-n) = 0$.

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	Largest 9621, smallest 1269 (1); Difference = 8352 (1).	1+1
22	Height = number ÷5 (1); Football: 7 cm; Chess: 3 cm; Hockey: 4 cm (1).	1+1
23	Try prime pairs (1); 5 + 19 = 24 (1).	1+1
24	Rectangle area = 60 cm^2 (1); Each triangle = 30 cm^2 (1).	1+1
25	LCM of 6 and 3 is 6 (1); $\frac{1}{2}$ (1).	1+1
26	Diagonals are equal and bisect each other (1); OA = OB = 6 cm; BD = 12 cm (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	Pieces = 3÷\frac{2}{5} (count $\frac{15}{2}$) (1); Whole pieces 7 (1); Left over $\frac{1}{5}$ m (1).	1+1+1
28	Diagonals of a square are perpendicular (1); $\angle AOB = 90°$ (1); Triangle AOB is isosceles right-angled so $\angle OAB = 45°$ (1).	1+1+1
29	(i) Vertical line: yes (1); (ii) Horizontal line: no (1); (iii) Rotational symmetry of order 2: no (1).	1+1+1
30	At noon: $-4 + 9 = 5$°C (1); In the evening: $5 - 8 = -3$°C (1); Below 0°C (1).	1+1+1
31	Difference between terms is 2 (1); 10th term = $5 + 9 \times 2$ (add the difference 9 times) (1); = 23 (1).	1+1+1
	OR: One right angle is $90°$ (1); $180 \div 90 = 2$ (1); 3 right angles make $270°$ (1).	1+1+1
32	Digit sum = $7 \times 5 = 35$ (1); Not divisible by 3, because the digit sum is not a multiple of 3 (2).	1+2
	OR: A bar graph should fit on the page, so use the bigger scale (1); Largest value 60 gives 6 cm with 1 cm = 10 (1); Hence choose 1 cm = 10 (1).	1+1+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	Letters with vertical symmetry: M, A, T, H (1); Letters with horizontal symmetry: H (1); Order-2 rotational letters: H (1); Word symmetric (vertical): no (2).	1+1+1+2
34	Total $= 120 + (-80) + 50 + (-30) = 60$ (1); Best day: Mon (₹120) (1); Worst day: Tue (₹-80) (1); Positive total means overall profit (2).	1+1+1+2
35	First person shakes 6 hands, next 5 new hands, ... down to 1 (1); Total $= 6 + 5 + 4 + 3 + 2 + 1$ (1); $= 21$ handshakes (1); This is the triangular number T_6 (2).	1+1+1+2
36	$\angle AOC = 135 - 45 = 90^\circ$ (1); $\angle AOB = 45^\circ$: acute (1); $\angle BOC = 135^\circ$: obtuse (1); $\angle AOC = 90^\circ$: right (1); State the arms and vertex (1).	1+1+1+1+1
	OR: Step 1: 8721-1278=7443 (1); Step 2: result 3996 (1); Step 3: result 6264 (1); The numbers move toward the Kaprekar constant 6174 (2).	1+1+1+2
37	Cricket: 4 symbols; Football: 5 symbols; Hockey: 3 symbols (1); Total symbols $= 12$ (1); Total students $= 60$ (1); Write the key: 1 symbol = 5 students (2).	1+1+1+2
	OR: Cross out multiples of 2, 3, 5, 7 (1); Primes: 41, 43, 47, 53, 59, 61, 67, 71, 73, 79 (1); Number of primes $= 10$ (1); Check by testing factors (2).	1+1+1+2
38	Area of first part $= 12 \times 5 = 60$ cm^2 (1); Area of second part $= 6 \times 4 = 24$ cm^2 (1); Total area $= 84$ cm^2 (1); Difference $= 36$ cm^2 (1); State the units (1).	1+1+1+1+1
	OR: Total =\frac{2}{5}+\frac{1}{4}= $13/20$ (1); Left $= 1 - 13/20 = 7/20$ (1); More on Monday (1); Check total plus left equals 1 (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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